

DIGITAL FORENSIC SECURITY

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DATA CARVING WITH XXD, BINWALK AND SCALPEL

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EXECUTIVE SUMMARY

This practical exercise involved the ability to explain why data carving is required when file-system metadata is missing or unreliable; use xxd to examine binary file content and identify headers and footers; recognise JPEG SOI and EOI signatures; reconstruct binary content from a hexadecimal dump; use The Sleuth Kit to examine inode, block and sector-level data; identify and extract embedded content using Binwalk; recover deleted files from a forensic USB image using Scalpel; verify recovered evidence using cryptographic hashes; and prepare a professional forensic report containing commands, screenshots, findings and limitations.

OBJECTIVES

The main objective of this project is to develop and demonstrate practical skills in digital forensic investigation through the structured examination of a forensic image in a controlled laboratory environment.

By completing this project, I aim to:

- Understand the importance of data carving when filesystem metadata is missing or unreliable.
- Examine binary files and identify file signatures using xxd.
- Recover and reconstruct deleted or fragmented files using forensic tools.
- Analyze inode, block, and sector-level data using The Sleuth Kit.
- Extract embedded content using Binwalk and recover deleted files using Scalpel.
- Verify recovered evidence using MD5 and SHA-256 hashes.

TOOLS USED AND RESOURCES

The tool to be used when it comes to offensive security lab project environment, we will be using several tools which are: Kali Linux OS, The sleuth Kit, MD5 & SHA- 256.

METHODOLOGY

In this methodology, I am going to solved the assignment using Phase, Part, step and we will be setting up our forensic workspace

Phase 1: Downloading our Forensic image file

In this phase, we will be creating different directory and download our original image file into working directory

```
(thony@kali)-[~/Downloads]
$ wget -q https://www.dropbox.com/s/nw23q14vzsykyp/Ch01InChap01.dd

(thony@kali)-[~]
$ ls
192.168.186.128  CIP-B101_Lab2A_OMOHUNLE_AKINWA  ctf1.txt  Documents  hello_with_newline.txt
192.168.186.2   CIP-B101_Lab1A_OMOHUNLE_AKINWA.tar.gz  ctf2      Downloads  hello_without_newline.txt
192.168.186.254 CIP-B101_Lab2A_OMOHUNLE_AKINWA.zip      ctf2.txt  evidence   ICDFA_Ch01InChap01_project
burp            CIP-B103_Lab2B_OMOHUNLE_AKINWA          Cyber_Project  evidence_text.txt  knock
Ch01InChap01.dd CIP-B102_Lab4_OMOHUNLE_AKINWA          Desktop    ForMitch.txt      knock-env
checkfileexit.sh CIP-B107_DataCarving                   Digital     greetings.sh       Music

(thony@kali)-[~]
$ cd CIP-B107_DataCarving/
Completing local directory
evidence/  hashes/  hex/  logs/  originals/  recovered/  report/  screenshots/  workings/
```

```
(thony@kali)-[~]
$ cp /home/thony/CIP-B107_DataCarving/originals/J_ub_law.jpg /home/thony/CIP-B107_DataCarving/workings

(thony@kali)-[~]
$ cp /home/thony/CIP-B107_DataCarving/originals/120M.7z /home/thony/CIP-B107_DataCarving/workings

(thony@kali)-[~]
$ cd CIP-B107_DataCarving/originals

(thony@kali)-[~/CIP-B107_DataCarving/originals]
$ ls
120M.7z  Ch01InChap01.dd  J_ub_law.jpg

(thony@kali)-[~/CIP-B107_DataCarving/originals]
$ cd /home/thony/CIP-B107_DataCarving/workings

(thony@kali)-[~/CIP-B107_DataCarving/workings]
$ ls
120M.7z  J_ub_law.jpg
```

Phase 2: Comparing original file hash and copy file using md5

In this phase, we will be making sure that our hash number are the same by comparing the original file hashing and the copied file hashing...

For copied file

```
(thony@kali) ~/CIP-B107_DataCarving/workings |
$ md5sum /home/thony/CIP-B107_DataCarving/workings/120M.7z
dfe7b5424e54cd1bf50d5df47aceeb3c /home/thony/CIP-B107_DataCarving/workings/120M.7z

(thony@kali) ~/CIP-B107_DataCarving/workings |
$ md5sum /home/thony/CIP-B107_DataCarving/workings/Ch01InChap01.dd
a117773bcf1fc88ec8ab8e0a349fbbcb /home/thony/CIP-B107_DataCarving/workings/Ch01InChap01.dd

(thony@kali) ~/CIP-B107_DataCarving/workings |
$ md5sum /home/thony/CIP-B107_DataCarving/workings/J_ub_law.jpg
83a360ac7f7e0ca318e5bfe39f95f137 /home/thony/CIP-B107_DataCarving/workings/J_ub_law.jpg

(thony@kali) ~/CIP-B107_DataCarving/workings |
$ sha256sum /home/thony/CIP-B107_DataCarving/workings/J_ub_law.jpg
238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468 /home/thony/CIP-B107_DataCarving/workings/J_ub_law.jpg

(thony@kali) ~/CIP-B107_DataCarving/workings |
$ sha256sum /home/thony/CIP-B107_DataCarving/workings/120M.7z
2d2a3d93c9ec65bcad9f89c9894429ea72caa8add07726a3b96ec9a6ab6a58ce /home/thony/CIP-B107_DataCarving/workings/120M.7z

(thony@kali) ~/CIP-B107_DataCarving/workings |
$ sha256sum /home/thony/CIP-B107_DataCarving/workings/Ch01InChap01.dd
3ce8053e4f3d9c8ab98b3aadb2480685efb8e4980d34297b83bd5a09b1a7b122 /home/thony/CIP-B107_DataCarving/workings/Ch01InChap01.dd
```

For original file

```
(thony@kali) ~/CIP-B107_DataCarving/workings |
$ sha256sum /home/thony/CIP-B107_DataCarving/originals/120M.7z
2d2a3d93c9ec65bcad9f89c9894429ea72caa8add07726a3b96ec9a6ab6a58ce /home/thony/CIP-B107_DataCarving/originals/120M.7z

(thony@kali) ~/CIP-B107_DataCarving/workings |
$ sha256sum /home/thony/CIP-B107_DataCarving/originals/Ch01InChap01.dd
3ce8053e4f3d9c8ab98b3aadb2480685efb8e4980d34297b83bd5a09b1a7b122 /home/thony/CIP-B107_DataCarving/originals/Ch01InChap01.dd

(thony@kali) ~/CIP-B107_DataCarving/workings |
$ sha256sum /home/thony/CIP-B107_DataCarving/originals/J_ub_law.jpg
238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468 /home/thony/CIP-B107_DataCarving/originals/J_ub_law.jpg

(thony@kali) ~/CIP-B107_DataCarving/workings |
$ md5sum /home/thony/CIP-B107_DataCarving/originals/120M.7z
dfe7b5424e54cd1bf50d5df47aceeb3c /home/thony/CIP-B107_DataCarving/originals/120M.7z

(thony@kali) ~/CIP-B107_DataCarving/workings |
$ md5sum /home/thony/CIP-B107_DataCarving/originals/Ch01InChap01.dd
a117773bcf1fc88ec8ab8e0a349fbbcb /home/thony/CIP-B107_DataCarving/originals/Ch01InChap01.dd

(thony@kali) ~/CIP-B107_DataCarving/workings |
$ md5sum /home/thony/CIP-B107_DataCarving/originals/J_ub_law.jpg
83a360ac7f7e0ca318e5bfe39f95f137 /home/thony/CIP-B107_DataCarving/originals/J_ub_law.jpg
```

PART A

FILE SIGNATURES AND HEX ANALYSIS

Step 1A

In this part, we will be making use of J_ub_law.jpg. we will be showing permission of the file and we want to know, the file is originally a file or not

```
(thony@kali)-[~/CIP-B107_DataCarving/workings]
$ ls -lh /home/thony/CIP-B107_DataCarving/originals/J_ub_law.jpg
-rw-rw-r-- 1 thony thony 1.7M Sep  2 23:21 /home/thony/CIP-B107_DataCarving/originals/J_ub_law.jpg
```

```
(thony@kali)-[~/CIP-B107_DataCarving/workings]
$ file /home/thony/CIP-B107_DataCarving/originals/J_ub_law.jpg
/home/thony/CIP-B107_DataCarving/originals/J_ub_law.jpg: JPEG image data, Exif standard: [
rer=NIKON CORPORATION, model=NIKON D4, orientation=upper-left], baseline, precision 8, 192
```

Setp 2A: Copying J_ub_law.jpg file into hashes folder

```
(thony@kali)-[~]
$ md5sum /home/thony/CIP-B107_DataCarving/originals/J_ub_law.jpg | tee /home/thony/CIP-B
07_DataCarving/hashes/J_ub_law.jpg_original_md5.txt
3a360ac7f7e0ca318e5bfe39f95f137 /home/thony/CIP-B107_DataCarving/originals/J_ub_law.jpg

(thony@kali)-[~]
$ sha256sum /home/thony/CIP-B107_DataCarving/originals/J_ub_law.jpg | tee /home/thony/CI
P-B107_DataCarving/hashes/J_ub_law_original_sha256.txt
238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468 /home/thony/CIP-B107_Dat
Carving/originals/J_ub_law.jpg
```

Step 3A: Examine the JPEG with xxd

In this phase, we will be examining the xxd jpeg file using head and tail

```
(thony@kali)-[~/CIP-B107_DataCarving/workings]
$ xxd J_ub_law.jpg | head -n 2
00000000: ffd8 ffe1 17f0 4578 6966 0000 4949 2a00  ....Exif..II*.
00000010: 0800 0000 1000 0001 0300 0100 0000 4013  ....@..

(thony@kali)-[~/CIP-B107_DataCarving/workings]
$ xxd J_ub_law.jpg | tail -n 2
0019d1e0: ff00 5254 0938 040e 6ff5 f75a 9a54 f565  ..RT.8..o..Z.T.e
0019d1f0: d351 9ebf ffd9                .Q...
```

Step 4A: Create the plain hexadecimal dump

```
(thony@kali)~/CIP-B107_DataCarving/hex
$ xxd -p /home/thony/CIP-B107_DataCarving/workings/J_ub_law.jpg > J_ub_law_plain.hex

(thony@kali)~/CIP-B107_DataCarving/hex
$ ls
J_ub_law_plain.hex

(thony@kali)~/CIP-B107_DataCarving/hex
$ head J_ub_law_plain.hex
ffd8ffe117f045786966000049492a000800000010000001030001000000
401300000101030001000000d00c00000201030003000000ce000000601
03000100000002000000f01020012000000d40000001001020009000000
e60000001201030001000000010000001501030001000000030000001a01
0500010000000ef0000001b01050001000000f70000002801030001000000
020000003101020021000000ff0000003201020014000000200100009882
02000c000000340100009e9c010052000000400100006987040001000000
94010000440400000800080008004e494b4f4e20434f52504f524154494f
4e004e494b4f4e20443400c0c62d0010270000c0c62d001027000041646f
62652050686f746f73686f702032312e3120284d6163696e746f73682900

(thony@kali)~/CIP-B107_DataCarving/hex
$ tail J_ub_law_plain.hex -n 2
515af6763a80fc7f81df1c79f5e1dc0f53e3a4d55f45182ad52d2406abc1
ad6994790789d1e1ff0052540938040e6ff5f75a9a54f565d3519ebfffd9
```

Step A6: Reconstruct the JPEG

In this step, we will be using the hexadecimal dump to reconstruct a new JPEG

```
(thony@kali)~/CIP-B107_DataCarving
$ ls
evidence hashes hex logs originals recovered report screenshots working workings

(thony@kali)~/CIP-B107_DataCarving
$ rm working

(thony@kali)~/CIP-B107_DataCarving
$ xxd -r -p hex/J_ub_law_plain.hex recovered/J_ub_law_reconstructed.jpg

(thony@kali)~/CIP-B107_DataCarving
$ ls -lh recovered/J_ub_law_reconstructed.jpg
-rw-rw-r-- 1 thony thony 1.7M Sep  3 18:52 recovered/J_ub_law_reconstructed.jpg
```

Step A7: Compare the hashes

In this step, we will be comparing our reconstructed.jpg

```
(thony@kali)~/CIP-B107_DataCarving
$ md5sum recovered/J_ub_law_reconstructed.jpg
83a360ac7f7e0ca318e5bfe39f95f137  recovered/J_ub_law_reconstructed.jpg

(thony@kali)~/CIP-B107_DataCarving
$ sha256sum recovered/J_ub_law_reconstructed.jpg
238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468  recovered/J_ub_law_recon
structed.jpg
```


PART B

Examine Embedded Metadata-Like Content

Part B1: Examine visible strings with strings

In this phase, we will be examining visible strings

```
be.com/xap/1.0/mm/" xmlns:stRef="http://ns.adobe.com/xap/1.0/sType/ResourceRef#" xmlns:stEvt="http://ns.adobe.com/xap/1.0/sType/ResourceEvent#" xmlns:crs="http://ns.adobe.com/camera-raw-settings/1.0/" xmlns:lr="http://ns.adobe.com/lightroom/1.0/" aux:ApproximateFocusDistance="4294967295/1" aux:ImageNumber="115422" aux:Lens="17.0-35.0 mm f/2.8" aux:LensID="99" aux:LensInfo="170/10 350/10 28/10 28/10" aux:SerialNumber="2027750" dc:format="image/jpeg" photoshop:DateCreated="2013-10-08" photoshop:LegacyIPTCDigest="5E379AE71DB03222FA2DE651E47F460" photoshop:ColorMode="3" photoshop:ICCPProfile="Adobe RGB (1998)" xmp:CreateDate="2013-10-08T18:56:21" xmp:CreatorTool="NIKON D4 Ver.1.01" xmp:Label="Approved" xmp:MetadataDate="2020-08-20T10:20:05-04:00" xmp:ModifyDate="2020-08-20T10:20:05-04:00" xmp:Rating="0" xmpMM:DocumentID="adobe:docid:photoshop:307cd9db-3a87-6148-9874-c6730fd42952" xmpMM:InstanceID="xmp.iid:9e7f6b27-04e9-43e7-ae79-c3f2fb9c2ac9" xmpMM:OriginalDocumentID="B04B02E367C62737F8E78649729D6364" crs:AlreadyApplied="True" crs:AutoLateralCA="0" crs:AutoWhiteVersion="134348800" crs:Blacks2012="0" crs:BlueHue="0" crs:BlueSaturation="0" crs:CameraProfile="Embedded" crs:CameraProfileDigest="54650A341B5B5CCA8442D0B43A92BCE" crs:Clarity2012="0" crs:ColorNoiseReduction="0" crs:Contrast2012="0" crs:ConvertToGrayscale="False" crs:DefringeGreenAmount="0" crs:DefringeGreenHueHi="60" crs:DefringeGreenHueLo="40" crs:DefringePurpleAmount="0" crs:DefringePurpleHueHi="70" crs:DefringePurpleHueLo="30" crs:Exposure2012="0.00" crs:GrainAmount="0" crs:GreenHue="0" crs:GreenSaturation="0" crs:HasCrop="False" crs:HasSettings="True" crs:Highlights2012="0" crs:HueAdjustmentAqua="0" crs:HueAdjustmentBlue
```

```
(thony@kali) ~/CIP-B107_DataCarving/workings
$ xxd J_ub_law.jpg | head -n 30
00000000: ffd8 ffe1 17f0 4578 6966 0000 4949 2a00  ....Exif..II*.
00000010: 0800 0000 1000 0001 0300 0100 0000 4013  ....@.
00000020: 0000 0101 0300 0100 0000 d00c 0000 0201  ....
00000030: 0300 0300 0000 ce00 0000 0601 0300 0100  ....
00000040: 0000 0200 0000 0f01 0200 1200 0000 d400  ....
00000050: 0000 1001 0200 0900 0000 e600 0000 1201  ....
00000060: 0300 0100 0000 0100 0000 1501 0300 0100  ....
00000070: 0000 0300 0000 1a01 0500 0100 0000 ef00  ....
00000080: 0000 1b01 0500 0100 0000 f700 0000 2801  ....(
00000090: 0300 0100 0000 0200 0000 3101 0200 2100  ....1...!.
000000a0: 0000 ff00 0000 3201 0200 1400 0000 2001  ....2.....
000000b0: 0000 9882 0200 0c00 0000 3401 0000 9e9c  ....4....
000000c0: 0100 5200 0000 4001 0000 6987 0400 0100  ..R...@...i....
000000d0: 0000 9401 0000 4404 0000 0800 0800 0800  ....D.....
000000e0: 4e49 4b4f 4e20 434f 5250 4f52 4154 494f  NIKON CORPORATIO
000000f0: 4e00 4e49 4b4f 4e20 4434 00c0 c62d 0010  N.NIKON D4...-..
00000100: 2700 00c0 c62d 0010 2700 0041 646f 6265  '....-..'..Adobe
00000110: 2050 686f 746f 7368 6f70 2032 312e 3120  Photoshop 21.1
00000120: 284d 6163 696e 746f 7368 2900 3230 3230  (Macintosh).2020
00000130: 3a30 383a 3230 2031 303a 3230 3a30 3500  :08:20 10:20:05.
00000140: 484f 5741 5244 204b 4f52 4e00 3200 3000  HOWARD KORN.2.0.
00000150: 3100 3300 2c00 2000 6300 6100 6d00 7000  1.3,.. .c.a.m.p.
00000160: 7500 7300 2c00 2000 4d00 6500 7200 7200  u.s,.. .M.e.r.r.
00000170: 6900 6300 6b00 2000 5300 6300 6800 6f00  i.c.k. .S.c.h.o.
00000180: 6f00 6c00 2000 6f00 6600 2000 4200 7500  o.l. .o.f. .B.u.
00000190: 7300 6900 6e00 6500 7300 7300 0000 0000  s.i.n.e.s.s....
000001a0: 2900 9a82 0500 0100 0000 8603 0000 9d82  ).....
000001b0: 0500 0100 0000 8e03 0000 2288 0300 0100  .....
000001c0: 0000 0100 0000 2788 0300 0100 0000 c800  .....
000001d0: 0000 3088 0300 0100 0000 0200 0000 0090  ..0.....
```

PART C

Direct Data-Unit Examination with The Sleuth Kit

Step C1: Confirm the forensic image

In this step, we will be confirming the image

```
(thony@kali)~/CIP-B107_DataCarving/workings
$ ls
120M,7z  Ch01InChap01.dd  J_ub_law.jpg

(thony@kali)~/CIP-B107_DataCarving/workings
$ ls -lh Ch01InChap01.dd
-rw-rw-r-- 1 thony thony 1.5M Sep  3 16:52 Ch01InChap01.dd
```

Step C2: Examine the image with img_stat

```
(thony@kali)~/CIP-B107_DataCarving/workings
$ ls
120M,7z  Ch01InChap01.dd  J_ub_law.jpg

(thony@kali)~/CIP-B107_DataCarving/workings
$ ls -lh Ch01InChap01.dd
-rw-rw-r-- 1 thony thony 1.5M Sep  3 16:52 Ch01InChap01.dd

(thony@kali)~/CIP-B107_DataCarving/workings
$ img_stat Ch01InChap01.dd
IMAGE FILE INFORMATION
-----
Image Type: raw

Size in bytes: 1474560
Sector size: 512

(thony@kali)~/CIP-B107_DataCarving/workings
$ img_stat Ch01InChap01.dd | tee /home/thony/CIP-B107_DataCarving/logs/part_c_img_stat.txt
IMAGE FILE INFORMATION
-----
Image Type: raw

Size in bytes: 1474560
Sector size: 512
```

Step C3: Examine the partition structure with mmls

```
(thony@kali)~/CIP-B107_DataCarving/workings
$ mmls Ch01InChap01.dd | tee /home/thony/CIP-B107_DataCarving/logs/part_c_mmls.txt
```

Step C4: Examine the filesystem with fsstat

```
(thony@kali)-[~/CIP-B107_DataCarving/workings]
$ fsstat Ch01InChap01.dd | tee /home/thony/CIP-B107_DataCarving/logs/part_c_fsstat.txt
FILE SYSTEM INFORMATION
-----
File System Type: FAT12

OEM Name: , 'k0nIHC
Volume ID: 0x0
Volume Label (Boot Sector):
Volume Label (Root Directory):
File System Type Label: ♦3♦♦M|

Sectors before file system: 0

File System Layout (in sectors)
Total Range: 0 - 2879
* Reserved: 0 - 0
** Boot Sector: 0
* FAT 0: 1 - 9
* FAT 1: 10 - 18
* Data Area: 19 - 2879
** Root Directory: 19 - 32
** Cluster Area: 33 - 2879

METADATA INFORMATION
-----
Range: 2 - 45782
Root Directory: 2

CONTENT INFORMATION
-----
Sector Size: 512
Cluster Size: 512
Total Cluster Range: 2 - 2848

FAT CONTENTS (in sectors)
-----
33-236 (204) → EOF
285-311 (27) → EOF
```

Step C5: List filesystem entries with fls

```
(thony@kali)-[~/CIP-B107_DataCarving/workings]
$ fls -r -p Ch01InChap01.dd | tee /home/thony/CIP-B107_DataCarving/logs/part_c_fls.txt
r/r 5: Client Info.mdb
r/r * 8: Billing Letter.doc
r/r * 11: confirmation.txt
r/r 13: Income.xls
r/r * 15: letter1.txt
r/r * 17: Regrets.doc
v/v 45779: $MBR
v/v 45780: $FAT1
v/v 45781: $FAT2
V/V 45782: $OrphanFiles
```

Step C6: Examine the actual inode with istat

```
(thony@kali)-[~/CIP-B107_DataCarving/workings]
$ istat Ch01InChap01.dd 13 | tee /home/thony/CIP-B107_DataCarving/logs/part_c_istat_13.txt
Directory Entry: 13
Allocated
File Attributes: File, Archive
Size: 13824
Name: INCOME.XLS

Directory Entry Times:
Written:      2005-12-09 06:52:18 (WAT)
Accessed:     2005-12-09 00:00:00 (WAT)
Created:      2005-12-09 06:59:07 (WAT)

Sectors:
285 286 287 288 289 290 291 292
293 294 295 296 297 298 299 300
301 302 303 304 305 306 307 308
309 310 311
```

Step C7: Examine the actual data unit with blkcat

```
(thony@kali)-[~/CIP-B107_DataCarving/workings]
$ blkcat Ch01InChap01.dd 285 | tee /home/thony/CIP-B107_DataCarving/logs/part_c_blkcat.txt
.....
```

Step C8: Recover the file with icat

```
(thony@kali)-[~/CIP-B107_DataCarving/workings]
$ icat Ch01InChap01.dd 13 > /home/thony/CIP-B107_DataCarving/recovered/Income.xls

(thony@kali)-[~/CIP-B107_DataCarving/workings]
$ file ../recovered/Income.xls
../recovered/Income.xls: Composite Document File V2 Document, Little Endian, Os: Windows,
Version 4.10, Code page: 1252, Author: Amelia Phillips, Last Saved By: Amelia Phillips, Name
of Creating Application: Microsoft Excel, Create Time/Date: Sat Nov 23 03:17:08 2002, Last
Saved Time/Date: Fri Dec 9 14:52:18 2005, Security: 0

(thony@kali)-[~/CIP-B107_DataCarving/workings]
$ ls -lh ../recovered/Income.xls
-rw-rw-r-- 1 thony thony 14K Sep  3 20:30 ../recovered/Income.xls
```

PART D

Embedded File Discovery with Binwalk

Step D1: Enter the working directory and checking, if Binwalk is installed

```
(thony@kali)~  
$ cd /home/thony/CIP-B107_DataCarving/workings  
  
(thony@kali)~/CIP-B107_DataCarving/workings  
$ ls  
120M,7z Ch01InChap01.dd J_ub_law.jpg  
  
(thony@kali)~/CIP-B107_DataCarving/workings  
$ binwalk
```

Step D4: Run Binwalk against the JPEG

```
(thony@kali)~/CIP-B107_DataCarving/workings  
$ binwalk J_ub_law.jpg | tee /home/thony/CIP-B107_DataCarving/logs/part_d_binwalk_jpeg.txt  
  
DECIMAL      HEXADECIMAL  DESCRIPTION  
-----  
0            0x0          JPEG image data, EXIF standard  
12           0xC          TIFF image data, little-endian offset of first image directory: 8  
26844       0x68DC       Copyright string: "Copyright 1999 Adobe Systems Incorporated"
```

Step D5: Run Binwalk against the forensic image

```
(thony@kali)~/CIP-B107_DataCarving/workings  
$ binwalk Ch01InChap01.dd | tee /home/thony/CIP-B107_DataCarving/logs/part_d_binwalk_dd.txt  
  
DECIMAL      HEXADECIMAL  DESCRIPTION  
-----
```


PART E

Prepare the USB Image

Step E1: Locate the archive

```
(thony@kali) - [~/CIP-B107_DataCarving/workings]
$ ls
120M.7z  Ch01InChap01.dd  J_ub_law.jpg
```

Step E2 — Record the archive hash

In this step, we will be recording hash before we unachieved the file

```
(thony@kali) - [~/CIP-B107_DataCarving/workings]
$ md5sum 120M.7z | tee /home/thony/CIP-B107_DataCarving/hashes/120M_Original_md5.txt
dfe7b5424e54cd1bf50d5df47aceeb3c 120M.7z

(thony@kali) - [~/CIP-B107_DataCarving/workings]
$ sha256sum 120M.7z | tee /home/thony/CIP-B107_DataCarving/hashes/120M_Original_sha256.txt
2d2a3d93c9ec65bcad9f89c9894429ea72caa8add07726a3b96ec9a6ab6a58ce 120M.7z
```

Step E3: Check 7-Zip

In this step, we will be listing the content of 7-Zip and then extract the file into our evidence directory

```
(thony@kali) - [~/CIP-B107_DataCarving/workings]
$ 7z l 120M.7z

7-Zip 26.02 (x64) : Copyright (c) 1999-2026 Igor Pavlov : 2026-06-25
64-bit locale=en_NG Threads:128 OPEN_MAX:4096, ASM

Scanning the drive for archives:
1 file, 36720470 bytes (36 MiB)

Listing archive: 120M.7z

--
Path = 120M.7z
Type = 7z
Physical Size = 36720470
Headers Size = 214
Method = LZMA2:24
Solid = +
Blocks = 1

  Date       Time       Attr      Size  Compressed  Name
-----
2021-09-23  03:59:31  D...A      0        0      120M
2021-09-23  03:59:31  ....A  124780544   36720256  120M/usb_fat_carving.001
2021-09-23  03:59:32  ....A    1685        120M/usb_fat_carving.001.txt
-----
2021-09-23  03:59:32          124782229   36720256  2 files, 1 folders
```

```

(thony@kali)~/CIP-B107_DataCarving
$ 7z x /home/thony/CIP-B107_DataCarving/workings/120M.7z -o../CIP-B107_DataCarving/evidence/120M_extracted

7-Zip 26.02 (x64) : Copyright (c) 1999-2026 Igor Pavlov : 2026-06-25
64-bit locale=en_NG Threads:128 OPEN_MAX:4096, ASM

Scanning the drive for archives:
1 file, 36720470 bytes (36 MiB)

Extracting archive: /home/thony/CIP-B107_DataCarving/workings/120M.7z
--
Path = /home/thony/CIP-B107_DataCarving/workings/120M.7z
Type = 7z
Physical Size = 36720470
Headers Size = 214
Method = LZMA2:24
Solid = +
Blocks = 1

Everything is Ok

Folders: 1
Files: 2
Size:      124782229
Compressed: 36720470

```

Step E4: Identify the extracted USB image

```

(thony@kali)~/CIP-B107_DataCarving
$ find ../CIP-B107_DataCarving/evidence/120M_extracted -type f -exec file {} \;
../CIP-B107_DataCarving/evidence/120M_extracted/120M/usb_fat_carving.001.txt: Unicode text
, UTF-8 (with BOM) text, with CRLF line terminators
../CIP-B107_DataCarving/evidence/120M_extracted/120M/usb_fat_carving.001: DOS/MBR boot sec
tor MS-MBR Windows 7 english at offset 0x163 "Invalid partition table" at offset 0x17b "Er
ror loading operating system" at offset 0x19a "Missing operating system", disk signature 0
xa1159a00; partition 1 : ID=0xe, active, start-CHS (0x0,2,3), end-CHS (0xe,254,63), starts
ector 128, 243584 sectors

(thony@kali)~/CIP-B107_DataCarving
$ find ../CIP-B107_DataCarving/evidence/120M_extracted
../CIP-B107_DataCarving/evidence/120M_extracted
../CIP-B107_DataCarving/evidence/120M_extracted/120M
../CIP-B107_DataCarving/evidence/120M_extracted/120M/usb_fat_carving.001.txt
../CIP-B107_DataCarving/evidence/120M_extracted/120M/usb_fat_carving.001

(thony@kali)~/CIP-B107_DataCarving
$ find ../CIP-B107_DataCarving/evidence/120M_extracted -type f -ls
2519345      4 -rw-rw-r--  1 thony  thony    1685 Sep 23  2021 ../CIP-B107_DataCa
rving/evidence/120M_extracted/120M/usb_fat_carving.001.txt
2519342 121856 -rw-rw-r--  1 thony  thony   124780544 Sep 23  2021 ../CIP-B107_DataC
arving/evidence/120M_extracted/120M/usb_fat_carving.001

```

Step E6: Copying and Recording USB image size

```

(thony@kali)~/CIP-B107_DataCarving
$ cp evidence/120M_extracted/120M/usb_fat_carving.001 ../CIP-B107_DataCarving/evidence

(thony@kali)~/CIP-B107_DataCarving
$ ls -lh evidence/usb_fat_carving.001
-rw-rw-r-- 1 thony thony 119M Sep  4 10:13 evidence/usb_fat_carving.001

(thony@kali)~/CIP-B107_DataCarving
$ cp evidence/120M_extracted/120M/usb_fat_carving.001.txt ../CIP-B107_DataCarving/evidence

(thony@kali)~/CIP-B107_DataCarving
$ ls -lh evidence/usb_fat_carving.001.txt
-rw-rw-r-- 1 thony thony 1.7K Sep  4 10:14 evidence/usb_fat_carving.001.txt

```

```
(thony@kali)~/CIP-B107_DataCarving
$ du -h evidence/usb_fat_carving.001
120M    evidence/usb_fat_carving.001

(thony@kali)~/CIP-B107_DataCarving
$ du -h evidence/usb_fat_carving.001.txt
4.0K    evidence/usb_fat_carving.001.txt
```

Step E7: Calculate USB image hashes

```
(thony@kali)~/CIP-B107_DataCarving
$ md5sum evidence/usb_fat_carving.001 | tee ../CIP-B107_DataCarving/hashes/usb_image_md5.txt
ba4a1d0ba49f4a6667b00a3b3e85e604  evidence/usb_fat_carving.001

(thony@kali)~/CIP-B107_DataCarving
$ sha256sum evidence/usb_fat_carving.001 | tee ../CIP-B107_DataCarving/hashes/usb_image_sha256.txt
zsh: no such file or directory: ../CIP-B107_DataCarving/hashes/usb_image_sha256.txt

(thony@kali)~/CIP-B107_DataCarving
$ sha256sum evidence/usb_fat_carving.001 | tee ../CIP-B107_DataCarving/hashes/usb_image_sha256.txt
9bfe4b5634ade30764f0e581a4686930f7fab0472378595b789b0cdb248c91d9  evidence/usb_fat_carving.001
```

Step E8: Examine the image with img_stat and copying it to log directory

```
(thony@kali)~/CIP-B107_DataCarving
$ img_stat evidence/usb_fat_carving.001 | tee ../CIP-B107_DataCarving/logs/part_e_img_stat.txt
IMAGE FILE INFORMATION
-----
Image Type: raw

Size in bytes: 124780544
Sector size: 512
```

Step E9: Confirm the partition offset with mmls and copying it to log directory

```
(thony@kali)~/CIP-B107_DataCarving
$ mmls evidence/usb_fat_carving.001 | tee ../CIP-B107_DataCarving/logs/part_e_mmls.txt
DOS Partition Table
Offset Sector: 0
Units are in 512-byte sectors

  Slot      Start      End      Length    Description
000:  Meta      0000000000  0000000000  0000000001  Primary Table (#0)
001:  _____ 0000000000  0000000127  0000000128  Unallocated
002:  000:000  0000000128  0000243711  0000243584  Win95 FAT16 (0x0e)
```


Step E10: Examine the filesystem

Since our mmls report gave us 128 at start output then now, we can examine our file using fsstat

```
(thony@kali) - [~/CIP-B107_DataCarving]
$ fsstat -o 128 evidence/usb_fat_carving.001 | tee ../CIP-B107_DataCarving/logs/part_e_f
sstat.txt
FILE SYSTEM INFORMATION
-----
File System Type: FAT16

OEM Name: MSDOS5.0
Volume ID: 0x94105672
Volume Label (Boot Sector): NO NAME
Volume Label (Root Directory): USB
File System Type Label: FAT16

Sectors before file system: 128

File System Layout (in sectors)
Total Range: 0 - 243583
* Reserved: 0 - 3
** Boot Sector: 0
* FAT 0: 4 - 241
* FAT 1: 242 - 479
* Data Area: 480 - 243583
** Root Directory: 480 - 511
** Cluster Area: 512 - 243583

METADATA INFORMATION
-----
Range: 2 - 3889670
Root Directory: 2

CONTENT INFORMATION
-----
Sector Size: 512
Cluster Size: 2048
Total Cluster Range: 2 - 60769

FAT CONTENTS (in sectors)
-----
512-515 (4) → EOF
516-519 (4) → EOF
520-523 (4) → EOF
70976-71211 (236) → EOF
71212-71215 (4) → EOF
```

PART F

CARVE DELETED FILES WITH SCALPEL

Step F1: Check Scalpel

In this step, we will be checking our scalpel, backing our scalpel up and uncommenting the jpeg 2 jpeg file and save it.

```
(thony@kali) - [~/CIP-B107_DataCarving]
$ which scalpel
/usr/bin/scalpel

(thony@kali) - [~/CIP-B107_DataCarving]
$ sudo cp /etc/scalpel/scalpel.conf /etc/scalpel/scalpel.conf.bak
[sudo] password for thony:

(thony@kali) - [~/CIP-B107_DataCarving]
$ sudo nano /etc/scalpel/scalpel.conf
```

```
#
#      NONE      y      1000      FOREMOST
#
# -----
# GRAPHICS FILES
# -----
#
# AOL ART files
#   art      y      150000  \x4a\x47\x04\x0e      \xcf\xcb
#   art      y      150000  \x4a\x47\x03\x0e      \xd0\xcb\x00\x00
#
# GIF and JPG files (very common)
#   gif      y      5000000  \x47\x49\x46\x38\x37\x61      \x00\x3b
#   gif      y      5000000  \x47\x49\x46\x38\x39\x61      \x00\x3b
#   jpg      y      5242880  \xff\xd8\xff???Exif      \xff\xd9
#   jpg      y      5242880  \xff\xd8\xff???JFIF      \xff\xd9
#
#
```

Step F4: Create a dedicated carving output directory and run scalpel

In this step, we will be creating a directory called scalpel and we will be running our scalpel command, after that we will save the output into scalpel directory

```
(thony@kali)~[~/CIP-B107_DataCarving]
$ mkdir recovered/scalpel

(thony@kali)~[~/CIP-B107_DataCarving]
$ sudo scalpel -c /etc/scalpel/scalpel.conf -o recovered/scalpel evidence/usb_fat_carvin
g.001
[sudo] password for thony:
Scalpel version 1.60
Written by Golden G. Richard III, based on Foremost 0.69.

Opening target "/home/thony/CIP-B107_DataCarving/evidence/usb_fat_carving.001"

Image file pass 1/2.
evidence/usb_fat_carving.001: 100.0% |*****| 119.0 MB 00:00 ETA
Allocating work queues...
Work queues allocation complete. Building carve lists...
Carve lists built. Workload:
jpg with header "\xff\xd8\xff\x3f\x3f\x3f\x45\x78\x69\x66" and footer "\xff\xd9" → 8 fil
es
jpg with header "\xff\xd8\xff\x3f\x3f\x3f\x4a\x46\x49\x46" and footer "\xff\xd9" → 9 fil
es
Carving files from image.
Image file pass 2/2.
evidence/usb_fat_carving.001: 100.0% |*****| 119.0 MB 00:00 ETA
Processing of image file complete. Cleaning up...
Done.
Scalpel is done, files carved = 17, elapsed = 1 seconds.
```

Step F6: Examine the Scalpel results

In this step, we will be examining the scalpel results

```
l-$ cat recovered/scalpel/audit.txt

Scalpel version 1.60 audit file
Started at Fri Sep 4 10:55:05 2026
Command line:
scalpel -c /etc/scalpel/scalpel.conf -o recovered/scalpel evidence/usb_fat_carving.001

Output directory: /home/thony/CIP-B107_DataCarving/recovered/scalpel
Configuration file: /etc/scalpel/scalpel.conf

Opening target "/home/thony/CIP-B107_DataCarving/evidence/usb_fat_carving.001"

The following files were carved:
File      Start      Chop      Length      Extracted From
00000010.jpg 3706992      NO      3148508      usb_fat_carving.00
1
00000009.jpg 425822      NO      4875802      usb_fat_carving.00
1
00000008.jpg 313872      NO      4965752      usb_fat_carving.00
1
00000002.jpg 6938624      NO      6868      usb_fat_carving.00
1
00000001.jpg 5285888      NO      1659604      usb_fat_carving.00
1
00000000.jpg 3897144      NO      3048148      usb_fat_carving.00
1
00000004.jpg 15427584      NO      4223822      usb_fat_carving.00
1
00000003.jpg 12881920      NO      4256310      usb_fat_carving.00
1
00000006.jpg 19638272      NO      5229050      usb_fat_carving.00
1
00000005.jpg 17121280      NO      4248530      usb_fat_carving.00
1
00000007.jpg 2135892      NO      5183267      usb_fat_carving.00
1
00000016.jpg 36671488      NO      2889262      usb_fat_carving.00
1
00000015.jpg 36571136      NO      2989614      usb_fat_carving.00
1
00000014.jpg 36393610      NO      3167148      usb_fat_carving.00
1
00000013.jpg 36332148      NO      3208302      usb_fat_carving.00
```

Step F7: Check recovered JPEG files

In this step, we will checking the recovered jpeg files and counting the number of images recovered

```
(thony@kali)~/CIP-B107_DataCarving
$ find recovered/scalpel -type f \( -iname "*.jpg" -o -iname "*.jpeg" \)
recovered/scalpel/jpg-1-0/00000012.jpg
recovered/scalpel/jpg-1-0/00000016.jpg
recovered/scalpel/jpg-1-0/00000015.jpg
recovered/scalpel/jpg-1-0/00000011.jpg
recovered/scalpel/jpg-1-0/00000009.jpg
recovered/scalpel/jpg-1-0/00000014.jpg
recovered/scalpel/jpg-1-0/00000013.jpg
recovered/scalpel/jpg-1-0/00000010.jpg
recovered/scalpel/jpg-1-0/00000008.jpg
recovered/scalpel/jpg-0-0/00000005.jpg
recovered/scalpel/jpg-0-0/00000006.jpg
recovered/scalpel/jpg-0-0/00000004.jpg
recovered/scalpel/jpg-0-0/00000003.jpg
recovered/scalpel/jpg-0-0/00000001.jpg
recovered/scalpel/jpg-0-0/00000000.jpg
recovered/scalpel/jpg-0-0/00000002.jpg
recovered/scalpel/jpg-0-0/00000007.jpg

(thony@kali)~/CIP-B107_DataCarving
$ find recovered/scalpel -type f \( -iname "*.jpg" -o -iname "*.jpeg" \) | wc -l
17
```

Step F8 : Identify recovered file types

In this step, we will identifying whether the carved files are actually JPEG images rather than trusting the filename extension.

```
(thony@kali)~/CIP-B107_DataCarving
$ find recovered/scalpel -type f -exec file {} \;
recovered/scalpel/jpg-1-0/00000012.jpg: JPEG image data, JFIF standard 1.01, resolution (D
PI), density 120x120, segment length 16, comment: "Software: Snipaste", baseline, precisio
n 8, 604x158, components 3
recovered/scalpel/jpg-1-0/00000016.jpg: JPEG image data, JFIF standard 1.02, aspect ratio,
density 100x100, segment length 16, progressive, precision 8, 750x1334, components 3
recovered/scalpel/jpg-1-0/00000015.jpg: JPEG image data, JFIF standard 1.02, aspect ratio,
density 100x100, segment length 16, progressive, precision 8, 750x1334, components 3
recovered/scalpel/jpg-1-0/00000011.jpg: JPEG image data, JFIF standard 1.01, resolution (D
PI), density 120x120, segment length 16, comment: "Software: Snipaste", baseline, precisio
n 8, 1667x762, components 3
recovered/scalpel/jpg-1-0/00000009.jpg: JPEG image data, JFIF standard 1.01, resolution (D
PI), density 220x220, segment length 16, baseline, precision 8, 360x540, components 3
recovered/scalpel/jpg-1-0/00000014.jpg: JPEG image data, JFIF standard 1.01, resolution (D
PI), density 96x96, segment length 16, baseline, precision 8, 256x144, components 3
recovered/scalpel/jpg-1-0/00000013.jpg: JPEG image data, JFIF standard 1.01, resolution (D
PI), density 96x96, segment length 16, baseline, precision 8, 256x144, components 3
recovered/scalpel/jpg-1-0/00000010.jpg: JPEG image data, JFIF standard 1.01, aspect ratio,
density 1x1, segment length 16, progressive, precision 8, 618x927, components 3
recovered/scalpel/jpg-1-0/00000008.jpg: JPEG image data, JFIF standard 1.01, resolution (D
PI), density 220x220, segment length 16, baseline, precision 8, 360x540, components 3
recovered/scalpel/audit.txt: ASCII text
```


Step F9: Hash the recovered JPEGs

MD5SUM

```
(thony@kali)~/CIP-B107_DataCarving
$ find recovered/scalpel -type f \( -iname "*.jpg" -o -iname "*.jpeg" \) -exec md5sum {} \; | tee ../CIP-B107_DataCarving/hashes/scalpel_jpeg_md5.txt
aec4abf7cb00ca5fcf54967d454a801d recovered/scalpel/jpg-1-0/00000012.jpg
fd36a3452eef212a0d64c824c2c18f9 recovered/scalpel/jpg-1-0/00000016.jpg
476bca75da4071d82cc636dfbad9cfc2 recovered/scalpel/jpg-1-0/00000015.jpg
a6eb06fcd32714f387cea59f8770b0c1 recovered/scalpel/jpg-1-0/00000011.jpg
e56584e188ed1852d33963cbc88478d4 recovered/scalpel/jpg-1-0/00000009.jpg
1dcdc61ee62aba1a876b57337df9c74 recovered/scalpel/jpg-1-0/00000014.jpg
160635e7a4dbc89384864bb5a9b6d952 recovered/scalpel/jpg-1-0/00000013.jpg
41cb821d0e0657ed0d2fab09d685dfc2 recovered/scalpel/jpg-1-0/00000010.jpg
6ea7a6c2e3988a8147e58cae15dc9d6 recovered/scalpel/jpg-1-0/00000008.jpg
8d6c3778abdb79fd36e87a79bd6abcc1 recovered/scalpel/jpg-0-0/00000005.jpg
23aebb41df892897e275e13145fead0 recovered/scalpel/jpg-0-0/00000006.jpg
85f5990b4cc0910d16c9a80b04f919a4 recovered/scalpel/jpg-0-0/00000004.jpg
d143fe24b792d62b55ad6b7e12c1dfa recovered/scalpel/jpg-0-0/00000003.jpg
e553e2ec5b6d2c67829a0edbcc7689cd recovered/scalpel/jpg-0-0/00000001.jpg
280d788eb3d1d48b93e88374440b1cc9 recovered/scalpel/jpg-0-0/00000000.jpg
4a5d891fd59cc27959925a6a93fb8ed8 recovered/scalpel/jpg-0-0/00000002.jpg
80d6a778d127d09b83f01dc8821b0907 recovered/scalpel/jpg-0-0/00000007.jpg
```

SHA256SUM

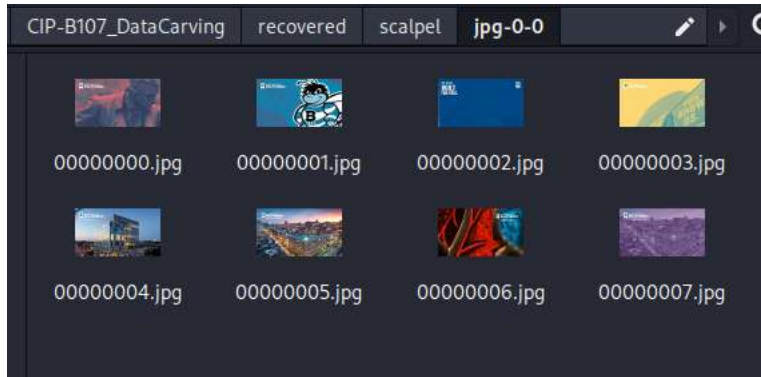
```
(thony@kali)~/CIP-B107_DataCarving
$ find recovered/scalpel -type f \( -iname "*.jpg" -o -iname "*.jpeg" \) -exec sha256sum {} \; | tee ../CIP-B107_DataCarving/hashes/scalpel_jpeg_sha256.txt
371807a9173648494a5ec4c3d63739ae6adeaab3646c15e29421eb64d46fbee4 recovered/scalpel/jpg-1-0/00000012.jpg
5b7a35a6b0bd67deaa1dbd92ee6b31950ace908c44f5124639d180a5e40e8567 recovered/scalpel/jpg-1-0/00000016.jpg
3d31fb0df55a4558446f8577cb7943a7dbd9fe44f782fdc8d64b13c4120d89d4 recovered/scalpel/jpg-1-0/00000015.jpg
dbf6aa689db8256e8d78cfb5d2016a66ee46dfc40690f9f1acbd8d467e673d4ee recovered/scalpel/jpg-1-0/00000011.jpg
5cf67210983c08d880e46cbd771b22b4a6bb0cccf96130b2106d6845094b06 recovered/scalpel/jpg-1-0/00000009.jpg
eb939a9b48da622896cf8577e9ccbd1c38a4b1686f09b3512282942d0c826411 recovered/scalpel/jpg-1-0/00000014.jpg
39a81569027ab59f0febe2bc518c6e519b763fb8ccd177b86841aaf84239b5a1 recovered/scalpel/jpg-1-0/00000013.jpg
7c127f3b90c3a0052cc019392ef4bb5ec17d7ec8a39fdd17d01ccd6a58be49ab recovered/scalpel/jpg-1-0/00000010.jpg
3eeeb18dbaaf50a6ae3cbb7630bd7fe426b317eb79cf951f3a6ceb601f8639d1 recovered/scalpel/jpg-1-0/00000008.jpg
8e49a4423aa2415b530b43b5d8ab9dbe72e1f6578e65ddd6e4202938be380e1 recovered/scalpel/jpg-0-0/00000005.jpg
0ff923813d2911727c7b72362f0d0990755a86f2bc6ca927ee8fcd0616a90573 recovered/scalpel/jpg-0-0/00000006.jpg
19b2e1bf7cdb40192d73157a1c70af5cfd89959125416c0a21a6afcd2ffdf384e recovered/scalpel/jpg-0-0/00000004.jpg
a8c87f31f3876d99110dd91a071697a424ac7f02aca4607941f7dacc26bbc47f recovered/scalpel/jpg-0-0/00000003.jpg
a3f54f4e312899bcd60f31fe99a3c392b3b2c46555c993f1d22581980073be33 recovered/scalpel/jpg-0-0/00000001.jpg
df36e6bcb12c782ab67a6662b5d8d306383cfe67ca050485a0f97d2db1aff293 recovered/scalpel/jpg-0-0/00000000.jpg
15e20d485aa30904f7b665c2ff297e8a9b01d0fc30b8dd0b4ffe9a47f4376a18 recovered/scalpel/jpg-0-0/00000002.jpg
9b8a2249a0ecbd7abfdac4d98cd031f61c4e740312a09ba3f2d132d1c04bae62 recovered/scalpel/jpg-0-0/00000007.jpg
```

Step F10: Verify at least two recovered JPEGs

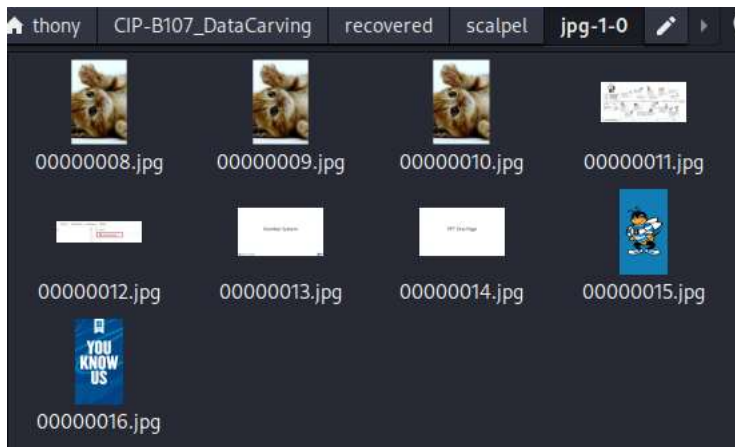
```
(thony@kali) - [~/CIP-B107_DataCarving]
$ ls recovered/scalpel/
audit.txt  jpg-0-0  jpg-1-0
```

```
(thony@kali) - [~/CIP-B107_DataCarving]
$ xdg-open recovered/scalpel/jpg-0-0
```

First jpg-0-0



Second jpg-1-0



SCREENSHOT AND METHODOLOGY

In this assignment we are going to solved it by exercises and everything is answer in methodology:

ANALYSIS AND FINDINGS

During my working on this project, I was able to gain real hands-on experience which involved the ability to explain why data carving is required when file-system metadata is missing or unreliable; The practical demonstrated how **xxd**, **strings**, **The Sleuth Kit**, **Binwalk**, and **Scalpel** can be used to examine raw data, identify file signatures, recover files, and analyze filesystem structures. MD5 and SHA-256 hashes were used to verify evidence integrity

Findings

- JPEG **SOI (FF D8)** and **EOI (FF D9)** signatures were identified.
- Hexadecimal data was examined and reconstructed.
- Metadata-like strings were searched using strings.
- Sleuth Kit was used to examine filesystem and data-unit information.
- Binwalk was used to identify embedded structures.
- Scalpel was used to carve recoverable JPEG files.
- Hashing was used to verify recovered evidence.

The original source image/device was not intentionally modified during forensic acquisition. The acquired image was stored separately from the source and used for subsequent examination.

CHALLENGES AND SOLUTIONS

In this chapter when working on my project, I experienced a challenge of starting what to do at first but at the end of the assignment I was able to solve and have much better experience in understanding it.

CONCLUSION

In this chapter, I was able to gain hand on experienced in understanding in digital forensics. Thanks to ICDFA, I was able to practically demonstrated **data carving can recover files when filesystem metadata is missing or unreliable**. The exercise also reinforced the importance of preserving original evidence, using verified forensic images, calculating hashes, and documenting every forensic step. These repeatable steps are useful in digital forensic which are also needed for any cybersecurity professional and vulnerable environments.